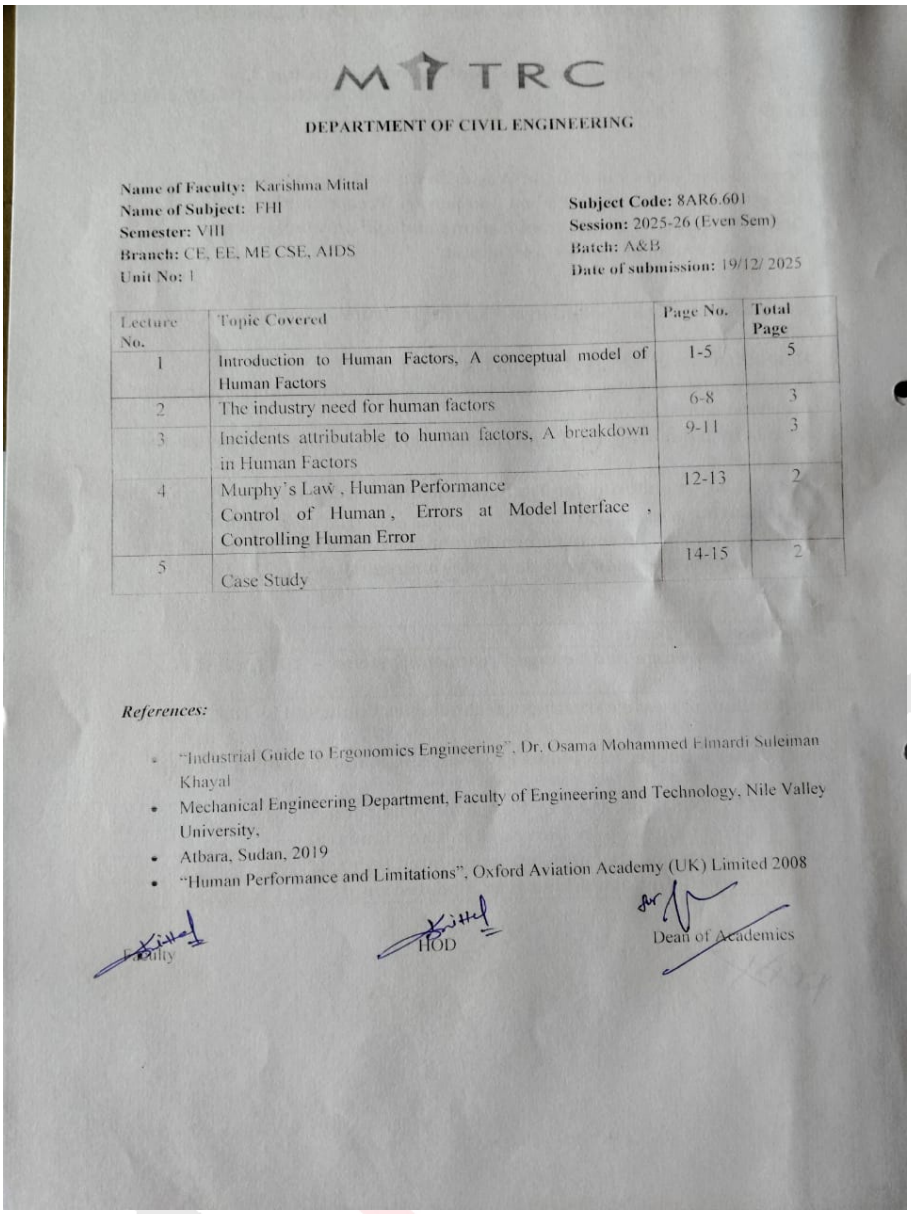


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DEPARTMENT OF CIVIL ENGINEERING

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Faculty HOD Dean of Academics

Lecture-1

Introduction to Human Factors

"Human factors" is the study of human capabilities and limitations in the workplace This information is applied to equipment, systems, facilities, procedures, and job training, for continual and effective human performance.

A Program Initially Designed by the Federal Aviation Administration (FAA). The Focus is on the Employees who Build and Maintain All Aircraft Engines and Aircraft.

Why Consider Human Factors?

- **Enhance Safety:** By minimizing errors and accidents caused by human limitations.
- **Improve Efficiency:** Aligning tools, processes, and systems with human abilities ensures faster and more accurate task completion.
- **Increase User Satisfaction:** User-friendly designs improve usability and experience.
- **Reduce Costs:** Preventing accidents and inefficiencies reduces financial and operational losses.
- **Support Adaptation:** Facilitates smoother integration of humans into complex systems.

A conceptual model of Human Factors/ Issues associated with Human Factor

The SHEL concept was first developed by Edwards in 1972, with a modified diagram to illustrate the model developed by Hawkins in 1975. The SHEL model can be used as an aid to understanding human factors.

- S - software
- H - hardware
- E - environment
- L – liveware

Human factors concentrates on the interfaces between the human (the 'L' at the centre of the model) and the other elements of the SHEL model

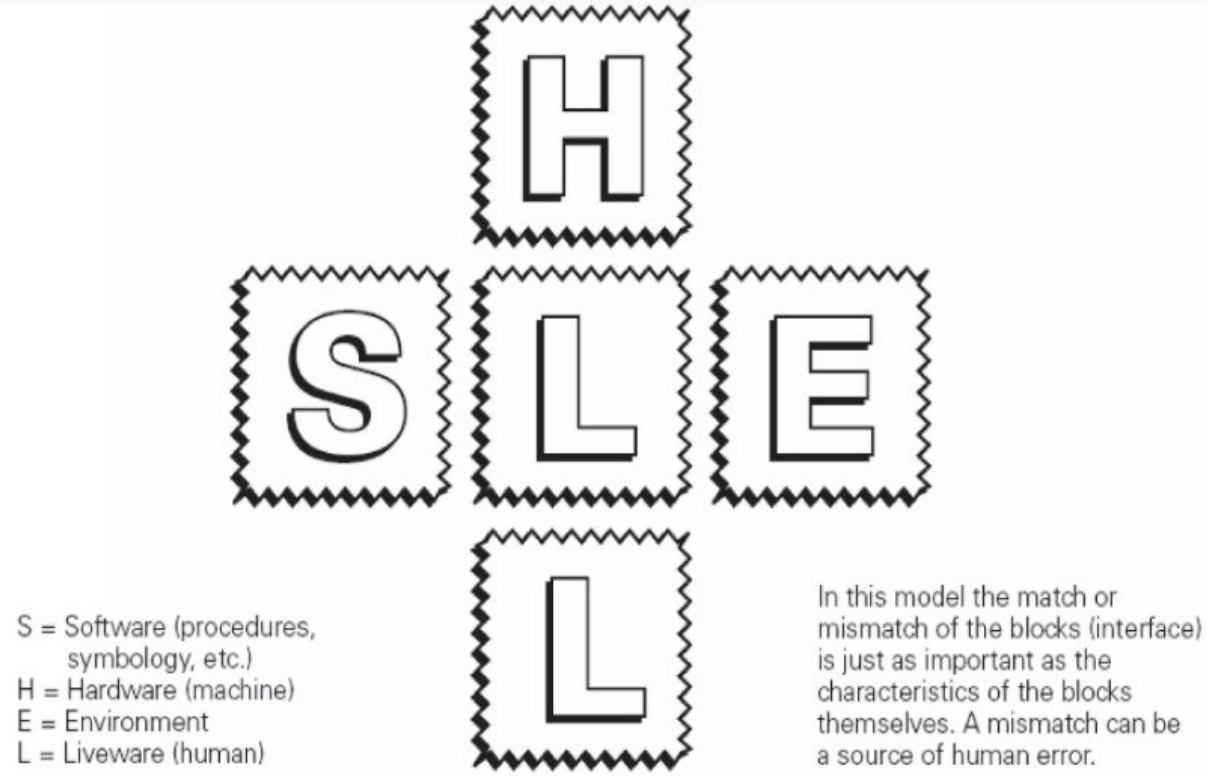


Fig-1 The SHEL Model

In the centre of the model is a person, the most critical as well as the most flexible component in the system. In order to achieve this matching, an understanding of the characteristics of this central component is essential. Some of the more important characteristics are the following:

1.**Physical Size and Shape**

- Body measurements and movements vary by age, ethnicity, and gender.
- Data from anthropometry and biomechanics guide workplace and equipment design.

2.**Physical Needs**

- Basic requirements include food, water, and oxygen.
- Insights are drawn from **physiology** and **biology**.

3. **Input Characteristics**

- Sensory systems collect external information but are prone to degradation.
- Knowledge from **physiology**, **psychology**, and **biology** is essential.

4.Information Processing

- Humans have limited memory and are affected by motivation and stress.
- Poor designs often ignore these limitations, highlighted by **psychology**.

5.Output Characteristics

- Processed information triggers muscle responses for actions or communication.
- Acceptable control parameters rely on **biomechanics**, **physiology**, and **psychology**.

The Liveware is the hub of the SHEL model of Human Factors. The remaining components must be adapted and matched to this central component.

Liveware-Hardware (L-H)

- Focuses on human-machine interaction (e.g., seat design, controls, displays).
- Humans often adapt to mismatches, masking deficiencies that may cause hazards.
- Modern challenges include advanced automation and computer integration.

Liveware-Software (L-S)

- Involves procedures, manuals, checklists, symbology, and software.
- Common issues include misinterpretation, non-compliance, and procedural errors.
- Problems are often hard to detect but frequently cited in accident reports.

Liveware-Environment (L-E)

- Initially adapted humans to the environment (e.g., helmets, anti-G suits).
- Now adapts the environment to human needs (e.g., pressurization, soundproofing).
- Modern concerns include radiation, disturbed sleep rhythms, and perceptual errors like illusions.
- Broader influences (e.g., political, economic) also affect this interface.

Liveware-Liveware (L-L)

- Concerns interpersonal interactions, including teamwork, leadership, and crew cooperation.
- Individual proficiency does not guarantee effective teamwork.
- Emphasizes group dynamics, corporate culture, and staff-management relationships.

Reference

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LECTURE-2

The industry need for human factors

The industry need for Human Factors is based on its impact on two broad areas, which interrelate so closely that in many cases their influences overlap and factors affecting one may also affect the other. These areas are:

• Effectiveness of the system

— safety

— efficiency

• Well-being of operational personnel

Effectiveness of the system

Safety

The best way to illustrate the effect of Human Factors issues on aviation safety is through the example of accidents. A few accidents in which aspects of Human Factors triggered the attention of the aviation community and paved the way to the proliferation of Human Factors endeavours in aviation are described here as examples.

1. — December 1972 — an L1011 crashed in the Florida Everglades (NTSB/AAR 73-14) and a B-737 crashed at Midway Airport in Chicago (NTSB/AAR 73- 16). In the first case, duties were not properly allocated and the whole flight crew became preoccupied with a landing gear indicator light bulb. In the second case, the captain — as a leader — did not properly manage the resources which were available to him
2. In 1974, a DC-10 crashed after take-off because a cargo door failed (it opened and blew out). The force applied by a cargo handler to close the cargo door, the door design and an incomplete application of a service bulletin were cited as factors (ICAO Circular 132- AN/93)

Efficiency

The need for application of Human Factors is not limited to flight safety. Efficiency is also radically influenced by the application of, or the lack of, Human Factors knowledge. For

instance, neglect of Human Factors in flight operations can be expected to cause less than optimum performance of tasks. Three of the many factors which may influence the well-being of operational personnel are **fatigue, body rhythm disturbance, and sleep deprivation or disturbance.** These are briefly explained below.

1.Fatigue

- Caused by inadequate rest or disturbed biological rhythms.
- **Acute Fatigue**: Results from long duty periods or demanding tasks in the short term.
- **Chronic Fatigue**: Accumulates over time due to prolonged stress or insufficient recovery.
- **Mental Fatigue**: Arises from emotional stress, even with physical rest.
- Contributing factors: **Hypoxia (low levels of oxygen in your body tissues)** and **noise**.
- Effects: Reduced safety, efficiency, and well-being.

2.Body Rhythm Disturbance

- **Circadian Rhythm**: A 24-hour cycle influenced by light, meals, and social activities.
- Disruptions impact safety, efficiency, and well-being, especially in long-haul and irregular flight schedules.
- Relevant for aircrew, air traffic controllers, and technicians on rotating shifts.
- **Jet Lag**: Symptoms include sleep disruption, irritability, anxiety, slowed reaction times, memory errors, and reduced performance.

3. Sleep

- Long-range flying disrupts normal sleep patterns, potentially causing sleep deprivation.
- Adults typically follow a single long sleep period daily, linked to brain rhythms.
- Sleep tolerance varies due to body chemistry and emotional stress.
- Effects include reduced alertness and operational effectiveness.

4.Stress in Aviation

1. **Sources of Stress**

- **Past**: Environmental factors like noise, vibration, temperature, and acceleration.
- **Present**:
  - Irregular working/resting patterns.
  - Disturbed circadian rhythms (e.g., long-range, night flights).
  - Life events (e.g., family separation, medical/proficiency checks, weddings).
  - High mental workload during critical phases (e.g., take-off, landing, emergencies).

2. **Impact on Performance**
   - Stress can impair decision-making, reaction time, and overall efficiency.
   - Responses to stress vary by individual; the same situation may be challenging for one person but overwhelming for another.

References

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Lecture-3

Incidents attributable to human factors/ human error-

In 1940, It was calculated that approximately 70% of all aircraft accidents were attributable to human error.

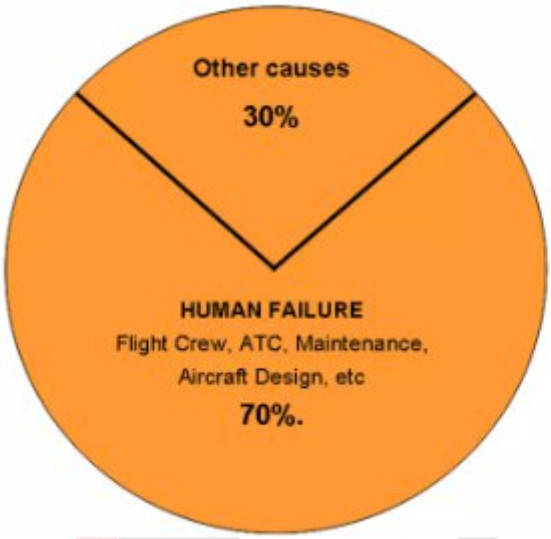


Fig-2 Survey of Human error

In 1986 a study was carried looking at significant accident causes in 93 aircraft accidents. These causes were as follows:

§ Pilot deviated from basic operational procedures - 33%

§ Inadequate cross-check by second crew member - 26%

§ Design faults -13%

§ Maintenance and inspection deficiencies - 12%

§ Absence of approach guidance - 10% As can be seen from the list, maintenance and inspection deficiencies are a major contributing factor to accidents

Aircraft maintenance and inspection duty can be very complex and varied in an environment where opportunities for error abound. Maintenance personnel frequently work under considerable time pressures.

Personnel at the maintenance base and at the flight line stations realize the importance of meeting scheduled departure times.

Operators have increased aircraft utilization in order to counteract the economic problems that plague the industry.

Human Error contributing

Human Error contributing factors Aircraft maintenance related incidents are characterised by:

§ Staff shortages

§ Time pressures

§ Shift or Task handovers

§ An element of a “can do” attitude

§ Errors occurred at night

§ Interruptions occurred

§ Some failure to use approved data or company procedures

§ Confusion using technical manuals

§ Inadequate pre-planning, equipment or spares

A breakdown in Human Factors

All accidents or incidents are avoidable if any of the number of steps could have been done differently.

Incidents and accidents can be prevented. By breaking any one of the links in the Error Chain.

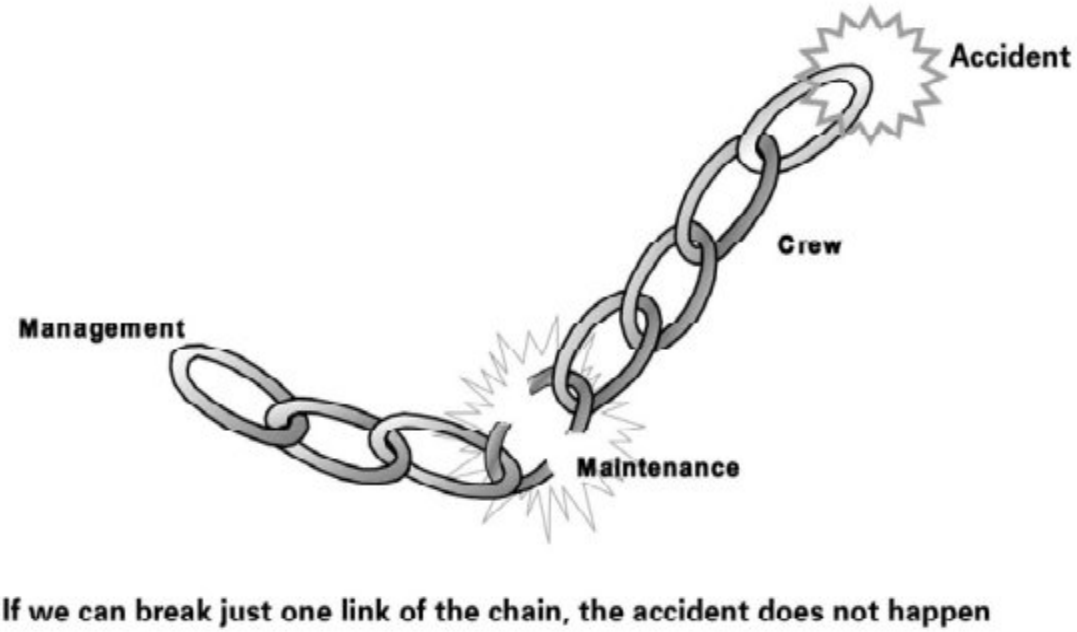


Fig-3 The Error Chain. Source

Breaking the Error Chain: An Example

Example: Wrong Runway Landing (Runway Incursion Accident)

Situation:
At a busy airport, an aircraft was about to land **on the wrong runway**.

Error Chain:

1. **Air Traffic Controller (ATC) Mistake** – The ATC mistakenly gave clearance **for the wrong runway**.
2. **Pilot Misinterpretation** – The pilot followed the instructions without reconfirming.
3. **Co-Pilot's Inaction** – The co-pilot did not double-check the runway number.
4. **Low Visibility** – Due to bad weather, the pilot could not clearly see the runway.

How Could the Error Chain Have Been Broken?

- **If the ATC had reconfirmed the runway clearance**, the mistake could have been avoided.
- **If the pilot had double-checked the instructions**, they could have identified the correct runway.
- **If the co-pilot had cross-checked the information**, the error could have been caught.
- **If an automatic alert system had warned the crew**, the landing could have been stopped in time.

References

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Lecture-4

Murphy's Law

IT can be regarded as the notion: "**If something can go wrong, it will."**

There is a tendency amongst human beings towards complacency. It is a belief that an accident will never happen to me or my company. It is important to convince individuals and organisations of the importance of human factor issues and how to:

§ Recognise risks and;

§ Implement Improvements

Examples of Murphy's Law

In Aviation:

- If a maintenance engineer **does not follow the safety checklist properly**, the chances of an error increase.
- If the pilot and co-pilot **do not cross-check information**, human error may lead to an accident.

In Daily Life:

- If there are two queues, **the one you choose will always move the slowest**.
- If you leave early for an important meeting, **you will definitely encounter heavy traffic**.

What Can We Learn from Murphy's Law?

1. **Plan Better** – Identify potential mistakes in advance and take preventive measures.
2. **Implement Safety Measures** – In aviation and engineering, always have backup systems in place to prevent failures.
3. **Stay Flexible** – Be prepared for unexpected problems by always having **a Plan B**

Human Performance and its signification in security system of Aviation

Just a certain mechanical component used in aircraft have limitations, engineers themselves have certain capabilities and limitations. Failures by aircraft maintenance engineers can also be to the detriment of aircraft safety

Control of Human

Error Some factors affecting Human Error:

- Carelessness
- Negligence or poor judgement
- Poorly designed equipment or may result from

- A normal reaction of a person to a particular situation

The latter kind of error is likely to be repeated and its occurrence can be anticipated. Kabul Afghanistan International Airport Fire Department

Errors at Model Interface

Each of the interfaces in the SHEL model has a potential of error where there is a mismatch between its components

Information Processing

- Before person can react to information, it must first be sensed; there is potential for error here because the sensory systems function within a narrow range.
- After conclusions have been formed about the meaning of a message, decision-making begins.
- Many factors may lead to erroneous decisions:

Example of poor Information Processing

Because of the high cost of aviation gasoline, a private Jet pilot once wrote to his aviation administration and asked if he could mix kerosene in his aircraft fuel. He received the following reply

"Regrettably decision involves uncertainties. Kerosene utilization consequences questionable, with respect to metalloferrous components and power production."

The Jet pilot did not process the information he had just received efficiently and replied:

"Thanks. This will sure cut my fuel bill."

He then received this urgent, and far clearer, communication from his aviation administration office

"DON'T USE KEROSENE. IT COULD KILL THE ENGINE – AND YOU TOO

Controlling Human Error

control of human error requires two different approaches:

1.Minimize occurrence of errors by:

- High levels of staff competence
- Design controls to match human characteristics
- Provide proper checklists, procedures, SOP's etc.
- Reduce stressful conditions

2.Reduce consequences of remaining errors by:

- Cross-monitoring Crews
- Close Crew co-operation

References

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Lecture-5

Case Studies

1. Tenerife Airport Disaster (1977) – Worst Aviation Accident in History

Location: Tenerife, Canary Islands, Spain
Aircraft Involved: KLM Flight 4805 & Pan Am Flight 1736 (Both Boeing 747s)
Fatalities: 583

Cause:

- Miscommunication between the KLM crew and air traffic control led the KLM pilot to take off while the Pan Am aircraft was still on the runway.
- The KLM captain, assuming he had clearance, initiated takeoff without receiving final confirmation.
- Dense fog reduced visibility, preventing both aircraft from seeing each other until it was too late.

Human Errors:

- Misinterpretation of air traffic control instructions.
- Failure of the Pan Am crew to vacate the runway in time.
- Poor cockpit communication and decision-making.

Lessons Learned:

- Improved standard phraseology in aviation communication.
- Mandatory Crew Resource Management (CRM) training for pilots.
- More structured and redundant communication between pilots and controllers.

2. Air France Flight 447 (2009) – Pilot Disorientation & Poor Decision-Making

Location: Atlantic Ocean
Aircraft: Airbus A330-203
Fatalities: 228

Cause:

- Icing caused the pitot tubes (which measure airspeed) to fail, leading to incorrect speed readings.
- The autopilot disconnected, and the pilots, unsure of the correct response, put the plane into a stall.
- Instead of lowering the nose to regain speed, the pilots kept pulling up, worsening the stall.

Human Errors:

- Failure to recognize and recover from a stall.
- Lack of experience in manual aircraft control at high altitudes.
- Poor coordination among the pilots in the cockpit.

Lessons Learned:

- Enhanced pilot training for manual flight recovery and high-altitude stall recovery.
- Improvements in aircraft sensor design to prevent similar failures.
- Emphasis on proper communication among crew members.

References

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